

COMPUTER ARCHITECTURE

20 TWO-MARK ANSWERS

1. Suitable architecture for an embedded system

Harvard architecture is suitable for many embedded systems because it has separate instruction and data memories.

It allows instruction fetch and data access simultaneously, improving speed and performance.

Example: Microcontrollers and DSPs.

2. Difference between CISC and RISC

CISC uses a large instruction set with complex instructions and many addressing modes.

RISC uses a smaller set of simple, regular instructions for faster execution and pipelining.

Example: x86 → CISC; ARM → RISC.

3. ARM Cortex and Intel x86

ARM Cortex processors are widely used in mobile and embedded systems due to low power consumption.

Intel x86 processors are mainly used in desktops, laptops and servers.

Example: Cortex-A → smartphones; Intel Core → desktops.

4. ARM instructions for addition and subtraction

ADD R0, R1, R2 ; $R0 = R1 + R2$

SUB R0, R1, R2 ; $R0 = R1 - R2$

ADD performs addition and **SUB** performs subtraction between two registers.

5. Major components of processor datapath

The major components are Register File, ALU, Multiplexers, Buses, PC (Program Counter) and IR (Instruction Register).

The ALU performs arithmetic/logic operations, while registers store operands and results.

6. Verilog statements for AND and OR

```
assign y_and = a & b;  
assign y_or = a | b;
```

& is the AND operator and **|** is the OR operator.
These statements describe combinational logic.

7. IEEE-754 fields

IEEE-754 single precision contains 32 bits:
Sign = 1 bit, Exponent = 8 bits, Mantissa/Fraction = 23 bits.
The exponent uses a bias of 127.

8. Purpose of microinstruction and execution stage

A microinstruction generates control signals for internal processor operations.
The execution stage performs the decoded instruction using the ALU, registers or memory.
Flow: Fetch → Decode → Execute.

9. Levels of memory hierarchy

The memory hierarchy is:
Registers → Cache → Main Memory (DRAM) → Secondary Storage (SSD/HDD).
From top to bottom, speed decreases while capacity increases.

10. Difference between SRAM, DRAM and Flash

SRAM: Fast, volatile, used in CPU cache.
DRAM: Slower, volatile, requires refresh, used as main memory.
Flash: Non-volatile, used in SSDs, USB drives and memory cards.

11. Define Instruction Set Architecture (ISA)

ISA is the interface between software and processor hardware.
It defines instructions, registers, data types, instruction formats and addressing modes.
Two design principles are simplicity and regularity.

12. What are addressing modes?

Addressing modes specify how an instruction identifies or obtains its operand.
Immediate: MOV R1, #10
Register: ADD R0, R1, R2

13. CPU execution time

Given: $IC = 2 \times 10^9$, $CPI = 2$, Clock Rate = 2 GHz.

CPU Time = $(IC \times CPI) / \text{Clock Rate}$

= $(2 \times 10^9 \times 2) / (2 \times 10^9) = 2$ seconds.

14. Purpose of FSM in processor control

An FSM (Finite State Machine) controls the sequence of processor operations.

It generates control signals for states such as Fetch → Decode → Execute → Write Back.

It is commonly used in the control unit.

15. Verilog statements for AND and OR

```
assign y = a & b; // AND
```

```
assign z = a | b; // OR
```

& performs AND, while | performs OR.

They are used to describe combinational circuits.

16. IEEE-754 single precision

IEEE-754 single precision has 32 bits divided into:

1-bit Sign + 8-bit Exponent + 23-bit Fraction.

The exponent has a bias of 127.

17. Purpose of microinstruction and execution stage

A microinstruction provides control signals for processor micro-operations.

The execution stage performs operations such as ADD, SUB, LOAD or STORE and produces the result.

18. Levels of memory hierarchy

Registers → L1/L2/L3 Cache → DRAM → SSD/HDD are the major levels.

Registers are fastest and smallest, while secondary storage is slowest and largest.

Cost per bit generally decreases down the hierarchy.

19. SRAM vs DRAM — speed and application

SRAM is faster and more expensive, so it is used for CPU cache memory.

DRAM is slower but cheaper and denser, so it is used for main memory (RAM).

Keyword: SRAM → Cache; DRAM → Main Memory.

20. SRAM, DRAM and Flash — key classification

SRAM → fastest → cache → volatile.

DRAM → slower → main memory → volatile + refresh required.

Flash → non-volatile → SSD/storage → retains data without power.